

A Lean 4 Certificate of Scott’s Measurement Structures and Linear Inequalities (1964) generated by LLM

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Abstract

This note records a Lean 4 / Mathlib formalization of Dana Scott’s 1964 paper *Measurement Structures and Linear Inequalities*. Scott gives a finite separation criterion for homogeneous linear inequalities and applies it to intransitive indifference, ordered utility differences, and qualitative probability. The development kernel-checks all eight numbered theorems (1.1–1.4, 2.1, 3.1, 3.2, 4.1) in about 4,200 lines across 21 modules. It also includes ordered-group remarks, signed-charge characterizations, literal vector cancellation, a Kraft–Pratt–Seidenberg counterexample to de Finetti’s axioms, and a separately labelled modern infinite reconstruction under an explicit generalized Kelley condition (not attributed to Scott’s unpublished announcement). There are no project axioms beyond Mathlib’s classical footprint [`propext`, `Quot.sound`, `Classical.choice`]; the `Scott1964/` library is sorry-free under `lake build`. Lean code was written by AI agents under the author’s direction and review. Proof summaries include short Lean fragments; the full library is indexed with hyperlinks to <https://github.com/catskillsresearch/scott1964>.

1 Introduction

Measurement theory asks when qualitative comparisons can be represented by real numbers. A relation on alternatives might be represented by utility differences, a relation on pairs might be represented by the sum of two utilities, and a relation on events might be represented by a finitely additive probability. The difficult direction is sufficiency: given only qualitative axioms, why must a numerical representation exist?

Scott’s paper gives a common answer. Encode each qualitative comparison as a linear inequality between vectors, then apply a finite separation theorem. For a finite symmetric set X in a real vector space and a chosen “nonnegative” part $N \subseteq X$, a realization is a linear functional φ satisfying

$$x \in N \iff 0 \leq \varphi(x) \quad (x \in X).$$

Sign completeness ensures that every $x \in X$ receives at least one weak sign. Cancellation prevents a positive combination of strictly positive vectors from collapsing into the neutral span. Together these conditions separate the strict cone from the neutral subspace and produce φ .

Rational coordinates permit positive real coefficients to be replaced by integer multiplicities, yielding Scott’s unweighted sequence conditions. The core predicates are as follows:

Lean 4 source

```
def Realizable (X N : Set L) : Prop :=
  exists phi : Module.Dual ? L, forall x in X, x in N <-> 0 <= phi x

def SignComplete (X N : Set L) : Prop :=
  forall ?x?, x in X -> x in N \ / -x in N

def WeightedSequenceCancellation (X N : Set L) : Prop :=
  forall (n : Nat) (x : Fin (n + 1) -> L) (c : Fin (n + 1) -> ?),
    (forall i, x i in X) -> (forall i, x i in N) -> (forall i, 0 < c i) ->
      ? i, c i * x i = 0 -> forall i, -x i in N
```

The three applications use different incidence vectors but the same engine.

1. **Intransitive indifference.** A strict relation P on a finite set is represented by $xPy \iff f(x) \geq f(y) + 1$.
2. **Ordered differences.** Comparisons of pairs are represented first by $f(x) + f'(x')$, then reversal combines the two scales into one utility difference $f(x) - f(y)$.
3. **Subjective probability.** An event in a finite Boolean algebra is sent to its 0/1 incidence vector on atoms. A separating functional gives atom weights, which normalize to a probability.

The formalization follows this mathematical decomposition rather than treating the eight theorems as unrelated endpoints. It also records where a proof route differs from Scott’s exposition: most importantly, Theorem 2.1 currently uses a direct finite Scott–Suppes staircase construction, while Scott’s local cycle reductions are formalized separately.

2 Scope and layout

The library has 21 Lean modules and approximately 4,200 lines. The root re-export module is `Scott1964/MeasurementStructures/Basic.lean`, which `Solution.lean` imports.

Module cluster	Role
<code>FinHead</code>	Stable distinguished index in <code>Fin (n + 1)</code> cancellation schemes
<code>LinearInequalities/Definitions, Separation, Sequences</code>	Geometric and sequence forms of Scott’s finite inequality criterion
<code>LinearInequalities/Rationalization, ScottTheorems</code>	Rational coefficient clearing and Theorems 1.1–1.4
<code>LinearInequalities/OrderedGroup</code>	Finite local real embedding and lexicographic obstruction
<code>Preference/Direct, Intransitive, Cycle</code>	Scott–Suppes construction, Theorem 2.1, and source proof reductions
<code>Differences/Pair, Ordered</code>	Incidence vectors and Theorems 3.1–3.2
<code>Probability/Basic, Atoms, Finite</code>	Finitely additive probabilities, atom vectors, signed charges, and Theorem 4.1
<code>Probability/KPSCounterexample</code>	Exact five-atom KPS qualitative order and nonrepresentability

Module cluster	Role
Probability/Infinite/*	Universal event space, Hahn–Banach/Kelley machinery, and the modern infinite reconstruction
Challenge / Solution	Optional stub module and sorry-free re-export of the library

The published scope is exactly the eight numbered theorems. Supplementary results are clearly separated from that inventory:

- `finite_local_real_embedding` formalizes Scott’s ordered-group observation;
- `scott_p15_signed_charge` and its vector form formalize the p. 15 signed-measure characterization;
- `deFinetti_axioms_insufficient` integrates the KPS five-atom obstruction;
- `reconstructed_infinite_theorem_4_1` is a modern theorem motivated by, but not identified with, Scott’s closing announcement.

The project is standalone. It imports none of the sibling formalizations of Scott’s later domain-theory papers.

3 How the proofs use mathlib

Mathlib supplies the finite-dimensional analytic and algebraic substrate. The predicates and reductions specific to Scott’s paper are defined in this repository.

Finite-dimensional separation. The key imported result is `geometric_hahn_banach_compact_closed` from mathlib’s locally convex separation library. The formalization specializes it to a finite convex hull and a finite-dimensional subspace. It then proves that the separator vanishes on the subspace and is strictly positive on the finite set:

Lean 4 source

```
theorem finite_strict_separation {A B : Set L} (hA : A.Finite) :
  Disjoint (convexHull ? A) (Submodule.span ? B : Set L) <->
    exists phi : Module.Dual ? L,
      (forall x in A, 0 < phi x) /\ forall x in B, phi x = 0
```

Finite combinatorics. Cancellation schemes are indexed by `Fin (n + 1)` so that a distinguished comparison always exists. `Equiv.Perm` expresses Scott’s two independent permutations in the pair and difference problems. `Finset` sums, subtype cardinalities, and `Equiv.ofFiber` turn equality of incidence-vector sums into actual permutations.

Linear algebra over function spaces. Scott’s vectors are represented as functions into $\mathbb{R}$. Pair incidence vectors live in $\text{Sum } A \ A' \rightarrow \mathbb{R}$; event vectors live in $\{a : B \mid \text{IsAtom } a\} \rightarrow \mathbb{R}$. Mathlib linear maps provide the representing functionals, while `Pi.single` and finite sums implement the coordinate encodings.

Finite Boolean algebras. Mathlib’s `BooleanAlgebra`, `IsAtom`, and finite-lattice API support decomposition of each event as the supremum of its atoms. The formalization proves both the atom-count and vector-sum versions of cancellation and constructs finitely additive functions by integrating atom weights.

Classical quotient orders. The direct proof of Theorem 2.1 uses `Antisymmetrization` of a finite total preorder. Substitutability classes remove duplicate preference profiles, and a recursively constructed staircase gives the numerical representation.

Functional analysis for the supplementary infinite theorem. The universal event space is a normed span of evaluation functions on all finitely additive probabilities. Continuous dual separation and a countable Kelley cover produce one functional that is nonnegative on weak comparisons and positive on strict comparisons.

What mathlib does not supply is Scott’s measurement-theoretic interface: the sign and cancellation conditions, relation-difference encoding, Scott–Suppes weak order, pair permutation principle, atom-vector transport, or generalized Kelley condition. Those constructions are developed here.

4 Proof dependency structure

The main published development follows one linear-inequality core and three applications.

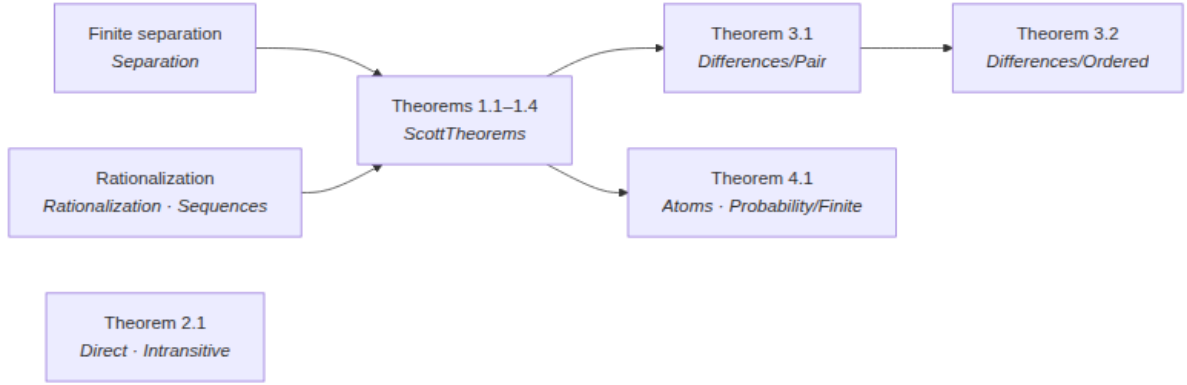


Figure 1: Main published dependency structure: finite separation core and the three applications.

Theorem 2.1 is shown as an independent branch because its completed proof is the direct Scott–Suppes construction. The source-facing cycle lemmas remain available for a future proof routed through the section-1 machinery.

The finite probability and modern infinite developments have distinct foundations:

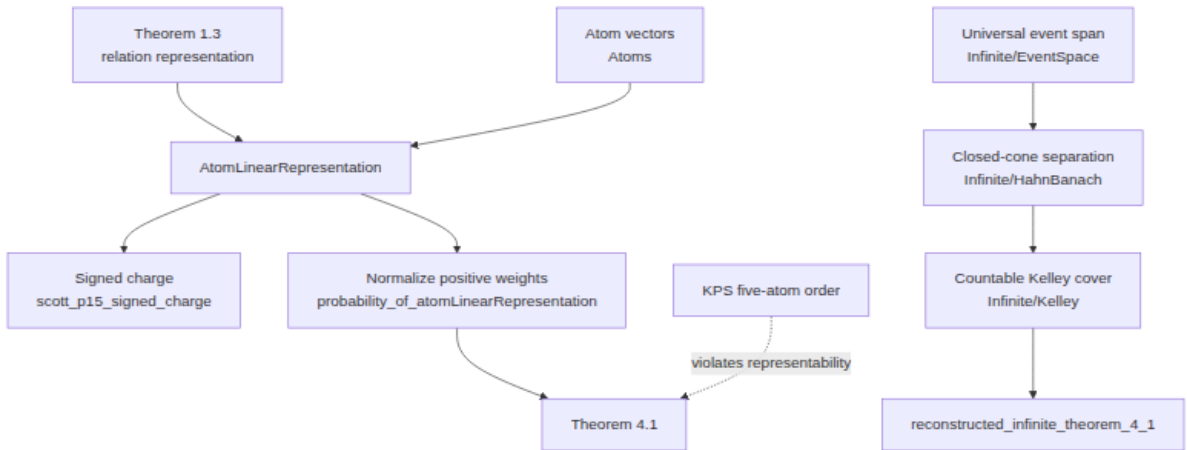


Figure 2: Finite subjective probability pipeline and the supplementary infinite reconstruction.

5 Theorem inventory

Paper result	Lean theorem	Content
Theorem 1.1	<code>LinearInequalities.scott_theorem_1_1</code>	Finite symmetric sign systems: realizability iff sign completeness and positive-weight cancellation
Theorem 1.2	<code>LinearInequalities.scott_theorem_1_2</code>	Rational-coordinate version with unweighted repeated summands
Theorem 1.3	<code>LinearInequalities.scott_theorem_1_3</code>	Complete relations on finite rational vectors: realizability iff paired equal-sums cancellation
Theorem 1.4	<code>LinearInequalities.scott_theorem_1_4</code>	Realizability iff extension to a strictly monotonic relation on the additive closure
Theorem 2.1	<code>theorem_2_1</code>	Intransitive-indifference representation with unit discrimination threshold
Theorem 3.1	<code>theorem_3_1</code>	Additive representation of pair comparisons by two utilities
Theorem 3.2	<code>theorem_3_2</code>	Ordered-difference representation by one utility
Theorem 4.1	<code>theorem_4_1</code>	Finite qualitative probability representation

Further source-facing or diagnostic results:

Topic	Lean theorem	Status
Literal form of (4_B)	<code>theorem_4_1_vector</code>	Equivalent vector-sum version of Theorem 4.1
Scott's p. 15 signed charge	<code>scott_p15_signed_charge</code> (+ vector form)	Proved without normalization or nonnegativity
Ordered-group remark	<code>finite_local_real_embedding</code>	Proved for each finite subset
Global obstruction	<code>no_global_real_additive_lex_realization</code>	Lexicographic $\mathbb{Z} \times \mathbb{Z}$ has no global additive real representation
KPS counterexample	<code>deFinetti_axioms_insufficient</code>	Exact five-atom order satisfies bundled de Finetti axioms but has no probability realization
Infinite analogue	<code>reconstructed_infinite_theorem_4_1</code>	Modern result under generalized Kelley condition; not attributed to Scott

6 Proof notes

6.1 Theorem 1.1 — finite separation

Let X be finite and symmetric, and let $N \subseteq X$ denote the vectors declared nonnegative. The strict vectors are those $x \in N$ for which $-x \notin N$; the neutral vectors have both signs. Scott's cancellation hypothesis states geometrically that the convex hull of the strict set is disjoint from the linear span of the neutral set.

`weightedCancellation_of_sequence` derives this geometric disjointness from the literal positive-weight sequence condition. It expands a hypothetical intersection point twice: as a convex combination of strict vectors and as a linear combination of neutral vectors. Signs of the latter coefficients are absorbed using symmetry, producing one positive finite dependence. Sequence cancellation would force the negative of a strict vector into N , a contradiction.

`finite_strict_separation` then produces a linear functional positive on strict vectors and zero on neutral vectors. Sign completeness proves that the functional realizes N :

Lean 4 source

```
theorem scott_theorem_1_1 {X N : Set L}
  (hX : X.Finite) (hsym : Symmetric X) :
  Realizable X N <->
    SignComplete X N /\ WeightedSequenceCancellation X N
```

The necessary direction is elementary: apply a realizing functional to a positive dependence and compare signs.

6.2 Theorem 1.2 — rationalization

Theorem 1.2 replaces arbitrary positive real coefficients with repeated unit coefficients when vectors have rational coordinates. The nontrivial direction of the rational weighted-to-unweighted reduction (`UnweightedSequenceCancellation.rationalWeighted`) is factored through `Rationalization.lean`:

The pipeline is: replace a positive real dependence by a positive rational dependence; clear denominators to positive natural multiplicities; repeat each vector according to its multiplicity; apply unweighted cancellation to the resulting finite sequence. This is the precise point where `[Fintype S]` and `IsRationalSet X` enter. Theorem 1.1 then supplies the realizing functional:

Lean 4 source

```
theorem scott_theorem_1_2 {S : Type*} [Fintype S] {X N : Set (S -> ?)}
  (hX : X.Finite) (hrat : IsRationalSet X) (hsym : Symmetric X) :
  Realizable X N <-> SignComplete X N /\ UnweightedSequenceCancellation X N
```

6.3 Theorems 1.3 and 1.4 — relations and additive closure

For a relation R on a finite set Y , Theorem 1.3 moves from points to differences $x - y$. Completeness gives sign completeness of the difference set. Scott's paired equal-sums condition gives unweighted cancellation:

$$\sum_i x_i = \sum_i y_i, \quad x_i R y_i \ (i \neq 0) \implies y_0 R x_0.$$

The proof constructs the symmetric rational difference set and applies Theorem 1.2. A short two-element cancellation argument ensures that the functional's weak inequality implies the original relation rather than only some relation with the same positive cone:

Lean 4 source

```
theorem scott_theorem_1_3 {S : Type*} [Fintype S]
  {Y : Set (S -> ?)} {R : (S -> ?) -> (S -> ?) -> Prop}
  (hY : Y.Finite) (hYrat : IsRationalSet Y) :
  RelationRealizable Y R <=>
    RelationComplete Y R /\ RelationSequenceCancellation Y R
```

Theorem 1.4 characterizes the same realizability by extension to the additive closure Y^+ . A realizing functional immediately defines the extension. Conversely, strict monotonicity gives paired cancellation by summing all comparisons except the distinguished one and cancelling equal totals:

Lean 4 source

```
theorem scott_theorem_1_4 {S : Type*} [Fintype S]
  {Y : Set (S -> ?)} {R : (S -> ?) -> (S -> ?) -> Prop}
  (hY : Y.Finite) (hYrat : IsRationalSet Y) :
  RelationRealizable Y R <=>
    exists Rplus,
      ExtendsOn Y (additiveClosure Y) R Rplus /\
      StrictlyMonotonic (additiveClosure Y) Rplus
```

6.4 Theorem 2.1 — intransitive indifference

Scott's relation permits indifference to be intransitive while preserving a real representation with a fixed discrimination threshold:

$$xPy \iff f(x) \geq f(y) + 1.$$

The forward implications to irreflexivity and the two quadruple axioms are linear arithmetic. The completed converse uses a direct finite Scott–Suppes construction. `ScottWeakOrder` compares alternatives by their strict-preference profiles. The quadruple axioms make it a total preorder; `Antisymmetrization` quotients alternatives with identical profiles. On the finite quotient, lower sections are nested, and `finite_staircase_representation` recursively assigns real values with the required unit gap. The representation is then pulled back to alternatives.

Lean 4 source

```
theorem theorem_2_1 {A : Type u} [Fintype A] [Nonempty A]
  (P : A -> A -> Prop) :
  RealizablePreference P <=>
    PrefIrrefl P /\ PrefQuadA P /\ PrefQuadB P
```

The library also verifies Scott's remarks around the theorem: every positive threshold is equivalent up to rescaling, every realization can be widened to a positive margin, and on a finite set the boundary can be avoided with a strict inequality. `Preference/Cycle.lean` separately proves transitivity and the three local cycle-shortening moves used in Scott's printed route. There is not yet a second completed proof assembling those reductions through the section-1 linear-inequality theorem.

6.5 Theorems 3.1 and 3.2 — pairs and differences

For Theorem 3.1, a pair (x, x') is represented by an incidence vector having one unit coordinate in each side of $\text{Sum } A \ A'$. Equality of sums of such vectors means that the first coordinates and second coordinates occur with the same multiplicities. `pairVector_sum_eq_permutations` upgrades those fiberwise cardinality equalities to two permutations.

The pair relation is transported to the finite rational range of `pairVector`. Pair totality becomes relation completeness and Scott’s permutation axiom becomes paired sequence cancellation. Theorem 1.3 then returns a functional φ . Restricting φ to the two kinds of singleton coordinates yields f and f' :

Lean 4 source

```
theorem theorem_3_1 {A : Type u} {A' : Type v}
  [Fintype A] [Nonempty A] [Fintype A'] [Nonempty A']
  (V : A -> A' -> A -> A' -> Prop) :
  RealizableUtilityPair V <-> PairTotal V /\ PairPermutation V
```

Theorem 3.2 treats a difference comparison $D(x, y, z, w)$ first as a pair comparison represented by $g(x) + q(y)$. Reversal supplies the second inequality needed to combine the scales. The function $f(x) = g(x) - q(x)$ then represents the ordered differences. Thus the sufficiency proof is a short application of Theorem 3.1 plus `difference_of_pair_and_reversal`.

Condition (2_D) remains an infinite scheme: Lean quantifies over every $n : \mathbb{N}$ and both permutations of `Fin (n + 1)`. No finite truncation is claimed.

6.6 Theorem 4.1 — finite subjective probability

For a finite Boolean algebra B , `atomVector` x is the real-valued characteristic function of the atoms below x . It is injective, and finite additivity is coordinate addition on disjoint suprema.

Scott writes (4_B) as equality of algebraic sums of characteristic functions. The primary `ProbCancellation` uses the equivalent atom-counting reading stated immediately after the theorem: each atom occurs below equally many events on the two sides. `ProbVectorCancellation` is the literal vector-sum form, and the equivalence is proved explicitly:

Lean 4 source

```
theorem probVectorCancellation_iff_probCancellation :
  ProbVectorCancellation R <-> ProbCancellation R
```

Totality and cancellation are transported through the range of `atomVector` and discharged by Theorem 1.3, yielding `AtomLinearRepresentation` R . Evaluation of the resulting functional on atom vectors is already a finitely additive signed charge. Nontriviality makes its value at $\top$ positive; nonnegativity makes all event values nonnegative. Division by the top value normalizes the charge without changing comparisons:

Lean 4 source

```
theorem theorem_4_1 {B : Type u} [BooleanAlgebra B] [Fintype B]
  (R : B -> B -> Prop) :
  RealizableProbability R <->
    ProbNontrivial R /\ ProbNonneg R /\
    ProbTotal R /\ ProbCancellation R
```

The structure `IsProbability` includes zero at $\perp$, normalization at $\top$, finite additivity on disjoint events, and nonnegativity. `IsSignedCharge` deliberately omits normalization and nonnegativity. This separation makes Scott’s p. 15 statement exact:

Lean 4 source

```
theorem scott_p15_signed_charge (R : B -> B -> Prop) :
  RealizableSignedCharge R <-> ProbTotal R /\ ProbCancellation R
```

Cancellation also derives reflexivity, transitivity, and both directions of de Finetti’s disjoint-union invariance. These lemmas connect Scott’s stronger cancellation criterion with the older qualitative-probability axioms.

6.7 Ordered groups and the lexicographic boundary

Scott observes after Theorem 1.4 that finite pieces of an ordered abelian group can be represented in the reals while preserving the addition equations visible in that piece. `finite_local_real_embedding` encodes order tests and local addition tests as rational coordinate vectors and applies Theorem 1.4. It produces an order-reflecting $f : S \rightarrow \mathbb{R}$ preserving every equation $x + y = z$ whose three terms lie in the finite set.

This local statement cannot be extended globally without an Archimedean hypothesis. The lexicographic group $\mathbb{Z} \times \mathbb{Z}$ supplies the counterexample. Its positive comparison is strictly monotonic on the additive closure of two generators, but any additive real representation would force one positive generator to dominate arbitrarily many multiples of the other. `no_global_real_additive_lex_realization` formalizes the contradiction.

6.8 The Kraft–Pratt–Seidenberg counterexample

`Probability/KPSCounterexample.lean` encodes the exact 32-event order on a five-atom powerset from Kraft, Pratt, and Seidenberg. Finite decidability is used to verify comparability, transitivity, bottom minimality, and de Finetti’s disjoint-union invariance. These are bundled as `DeFinettiAxioms`.

Nonrepresentability is not left to exhaustive computation. Finite additivity expresses every event weight as a sum of singleton weights, and selected strict comparisons in the KPS order force an impossible cycle of linear inequalities. Consequently:

Lean 4 source

```
theorem deFinetti_axioms_insufficient :  
  DeFinettiAxioms KPSGe /\ not RealizableProbability KPSGe
```

`DeFinettiAxioms` uses de Finetti’s literal weak nontriviality condition $\neg R \perp T$. Scott’s `ProbNontrivial` is stronger: $R \top \perp \wedge \neg R \perp T$.

6.9 Modern reconstruction of an infinite analogue

Scott’s final paragraph announces an extension to infinite Boolean algebras but gives neither a theorem statement nor hypotheses. The repository therefore does not attempt to recover or attribute that unpublished result. Instead it proves a modern analogue under a fully explicit condition.

`EventSpace.lean` embeds each event into the normed span of its evaluation function over all finitely additive probabilities. Stone-point indicators prove that this encoding is faithful. Weak comparisons generate a closed cone, described by its continuous-dual polar. Strict comparison vectors are covered by countably many closed convex upper sets avoiding zero—the `GeneralizedKelleyCondition`.

`Kelley.lean` separates each layer from zero and combines the separators with summable positive coefficients. The resulting continuous functional is nonnegative on every weak comparison and strictly positive on every strict comparison. Its value at T is positive, so normalization yields a finitely additive probability representing the relation:

Lean 4 source

```
theorem reconstructed_infinite_theorem_4_1 (R : B -> B -> Prop) :  
  RealizableProbability R <->  
    ProbNontrivial R /\ ProbNonneg R /\ ProbTotal R /\  
      GeneralizedKelleyCondition R
```

This theorem is a supplementary reconstruction, not a ninth theorem of Scott’s published paper.

7 Source fidelity

The working source is `sources/ScottMeasurement1964.pdf`. A searchable triple-pass vision transcription is generated at `sources/ScottMeasurement1964_vision.md` by `scripts/ocr_pdf_pipeline.sh`. The PDF and transcription are source material, not part of the Apache-2.0 grant.

7.1 Formal conventions

Scott writes finite sequences with an exceptional zeroth term. Lean uses `Fin (n + 1)` and the named index `finHead n`; this avoids empty-index edge cases and keeps cancellation statements uniform across modules.

Relations are oriented as “at least as preferred/probable.” Thus `RelationRealizable Y R` states $Rxy \leftrightarrow \varphi y \leq \varphi x$, and `RealizableProbability R` states $Rxy \leftrightarrow \mu(x) \geq \mu(y)$. Scott’s strict comparison is represented by `StrictlyPreferred R x y := ¬ R y x`.

The finite probability theorem uses abstract finite Boolean algebras rather than a concrete powerset. Characteristic vectors are indexed by Boolean atoms, which is equivalent to choosing the atoms of a finite field of events.

The development is classical. The disclosed axiom footprint is `propext`, `Quot.sound`, and `Classical.choice`; no project axiom is declared.

7.2 Recorded divergences and limitations

- Theorem 2.1 is completed by the direct finite Scott–Suppes staircase construction. Scott’s local cycle reductions are formalized in `Preference/Cycle.lean`, but they are not assembled into an alternate section-1 proof.
- The primary `theorem_4_1` uses Scott’s atom-count explanation of `(4_B). theorem_4_1_vector` separately proves the equivalent literal equality of characteristic-vector sums.
- Scott’s closing paragraph does not state his infinite theorem. `reconstructed_infinite_theorem_4_1` is explicitly a modern Hahn–Banach/Kelley reconstruction under `GeneralizedKelleyCondition`.
- The KPS development formalizes an external 1958 counterexample used to delimit the strength of de Finetti’s axioms; it is not a numbered theorem of Scott’s article.
- No external mathematical review has been performed. Every proof is checked by the Lean kernel, but the project remains self-assessed.

8 Build and preflight

The repository pins Lean and mathlib **v4.33.0**.

```
lake exe cache get
lake build
bash scripts/generate_arxiv_with_code.sh # -> arxiv_with_code.md
```

`lake build` checks the `Scott1964` library (and optional `Solution` re-export).

`arxiv_with_code.md` is generated (narrative plus Appendix A module index with GitHub links). Regenerate when `arxiv.md` or listed sources change; build the PDF and arXiv zip with `bash scripts/build_arxiv_pdf.sh`.

9 License and source PDF

Original Lean code and author-written documentation are Apache-2.0. `sources/ScottMeasurement1964.pdf` and its transcription are not Apache-2.0; see NOTICE and `sources/README.md` for the copyright carve-out.

10 Acknowledgments

The author thanks **Prof. Dana Scott** (Carnegie Mellon University) for suggesting *Measurement Structures and Linear Inequalities* as a formalization target. Scott did not participate in this work and does not endorse it.

10.1 AI-assisted development

Lean in this repository was written by AI agents under the author’s direction and review. The human author retains sole responsibility for the mathematical content, the formalization route, and every formal claim. **No large language model is listed as a co-author.**

We gratefully acknowledge assistance from the following tools:

- **Cursor [Cur26]** — agent-assisted editing in the Cursor IDE: formalizing Scott’s 1964 *Measurement Structures and Linear Inequalities* in Lean 4 / `mathlib`, `lake build` repair, vision-OCR transcription, drafting this narrative (`arxiv.md`), and tracking the formalized inventory. Generated Lean was provisional until it compiled under the pinned toolchain.
- **Cursor Composer 2.5 Fast [Cmp25]** — routine multi-step work: module scaffolding, dependency-ordered wiring of `Scott1964/MeasurementStructures/`, documentation, and medium proof obligations where the strategy was already fixed.
- **Anthropic Claude Sonnet 5 (medium reasoning) [Son26]** — day-to-day formalization and proof-engineering in Cursor at the medium reasoning tier: inventory and narrative maintenance, module wiring, and medium-complexity Lean obligations where the proof strategy was already fixed.
- **Anthropic Claude Opus 4.8 (high reasoning) [Ant26]** — selective use for the heaviest proof work. Every emitted proof term was checked by the Lean kernel.

List of Figures

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11 References

- [Cur26] Anysphere, Inc. *Cursor: AI-native code editor and agent environment*. <https://cursor.com> (accessed 2026).
- [Cmp25] Anysphere, Inc. *Composer 2.5*. Model announcement and documentation, <https://cursor.com/blog/composer-2-5>; model card as integrated in Cursor, <https://cursor.com/docs/models> (accessed 2026).
- [Son26] Anthropic. *Claude Sonnet 5* (medium reasoning variant). System card, <https://www.anthropic.com/claude-sonnet-5-system-card>; model documentation as integrated in Cursor, <https://cursor.com/docs/models> (accessed 2026).
- [Ant26] Anthropic. *Claude Opus 4.8* (high thinking/reasoning variant). System card and announcement, <https://www.anthropic.com/news/claude-opus-4-8>; model documentation as integrated in Cursor, <https://cursor.com/docs/models/claude-opus-4-8> (accessed 2026).

A Lean module index

Checked by lake build. Complete sources: <https://github.com/catskillsresearch/scott1964>. Each subsection links to the corresponding file on GitHub.

Role	File
Root import graph	Scott1964.lean
Stable distinguished sequence index	Scott1964/MeasurementStructures/FinHead.lean
Scott’s finite inequality predicates	Scott1964/MeasurementStructures/LinearInequalities/Definitions.lean
Finite convex separation	Scott1964/MeasurementStructures/LinearInequalities/Separation.lean
Rational coefficient reduction	Scott1964/MeasurementStructures/LinearInequalities/Rationalization.lean
Explicit finite sequence conditions	Scott1964/MeasurementStructures/LinearInequalities/Sequences.lean
Theorems 1.1–1.4	Scott1964/MeasurementStructures/LinearInequalities/ScottTheorems.lean
Ordered-group consequences and obstruction	Scott1964/MeasurementStructures/LinearInequalities/OrderedGroup.lean
Finite Scott–Suppes staircase representation	Scott1964/MeasurementStructures/Preference/Direct.lean
Local cycle reductions	Scott1964/MeasurementStructures/Preference/Cycle.lean
Theorem 2.1	Scott1964/MeasurementStructures/Preference/Intransitive.lean
Theorem 3.1	Scott1964/MeasurementStructures/Differences/Pair.lean
Theorem 3.2	Scott1964/MeasurementStructures/Differences/Ordered.lean

Role	File
Finite qualitative probability definitions	<code>Scott1964/MeasurementStructures/Probability/Basic.lean</code>
Atom-vector representation	<code>Scott1964/MeasurementStructures/Probability/Atoms.lean</code>
Theorem 4.1 and signed charges	<code>Scott1964/MeasurementStructures/Probability/Finite.lean</code>
Kraft–Pratt–Seidenberg counterexample	<code>Scott1964/MeasurementStructures/Probability/KPSCounterexample.lean</code>
Universal event space	<code>Scott1964/MeasurementStructures/Probability/Infinite/EventSpace.lean</code>
Closed-cone separation	<code>Scott1964/MeasurementStructures/Probability/Infinite/HahnBanach.lean</code>
Generalized Kelley cover	<code>Scott1964/MeasurementStructures/Probability/Infinite/Kelley.lean</code>
Modern infinite reconstruction	<code>Scott1964/MeasurementStructures/Probability/Infinite/Reconstructed.lean</code>
Complete library re-export	<code>Scott1964/MeasurementStructures/Basic.lean</code>

Primary source (PDF): [sources/ScottMeasurement1964.pdf](#) — Dana S. Scott, *Measurement Structures and Linear Inequalities* (J. Math. Psychology 1, 1964).

Total: 22 modules, 4182 lines of Lean.